

FschoolAI — Product Requirements Document (PRD)

Version: 1.2

Date: June 23, 2026

Author: Vincent Yang, FschoolAI

Audience: Engineering team — Tencent engineer, Bytedance engineer, Aryan, Ryan, Vincent

1. Product Vision

FschoolAI is the **personal academic intelligence** for every student. It is not a generic AI chatbot. It is not a homework helper. It is a persistent, living brain that knows each student individually — their learning style, their knowledge gaps, their deadlines, their stress level, their history — and uses that knowledge to help them from where they are to where they need to be.

The core principle is: **the AI adapts to the student, not the student to the AI.**

Every student is unique. A student at Carleton studying engineering has different gaps, different pressures, and different ways of learning than a student at UCLA studying pre-med. FschoolAI treats each student as a unique individual, not a user type.

At signup, a **student second brain** is created for each user. This brain is theirs. It grows with every session — every question asked, every assignment completed, every lecture attended. When NeuroAGI hardware launches, the student claims their brain on the device. Until then, the brain lives in the cloud and is accessible through FschoolAI.

2. Target Users

FschoolAI is built for university and college students globally, with initial focus on North America. The product must be immediately useful to an international student who is not a native English speaker, a first-generation university student with no academic support network, and a high-achieving student who wants to go beyond what their institution provides.

The product must be so useful that students are willing to pay for it personally — not wait for their school to buy it.

3. Core Architecture

3.1 The Student Brain

Every student has a brain object stored in the NeuroAGI brain layer. This is a persistent, structured knowledge graph about the student. It is the foundation that every agent reads from and writes to.

```
StudentBrain {
  student_id: string
  learning_style: "visual" | "auditory" | "reading" | "kinesthetic" | "mixed"
  knowledge_nodes: KnowledgeNode[] // what the student knows and gaps
  course_context: CourseContext[] // current courses, assignments,
  stress_level: float (0.0-1.0) // inferred from behaviour patterns
  session_history: Session[] // all past interactions
  performance_signals: Signal[] // quiz scores, time-on-task, completion
  preferences: Preferences // communication style, response time
  upcoming_deadlines: Deadline[] // from Canvas sync
  created_at: timestamp
  last_updated: timestamp
}
```

3.2 Agent Architecture

There are two distinct agent patterns. Using the wrong pattern for a given agent is a design error.

Pattern A — Request/Response (interactive agents) Used by: Reggie, Tutor, Canvas, Planner, Lecture, Library, Exam Mode, Audio, Office Hours, Calendar, Terminal.

```
Step 1: context = brain.read(student_id)
Step 2: result = agent_logic(user_input, context)
Step 3: brain.write(student_id, signal)
```

Pattern B — Watch/Arbitrate/Deliver (background/proactive agents) Used by: Intervention Agent, Reflection Agent, Cohort Agent. These agents do NOT wait for user input. They watch for events or run on schedule, evaluate whether an intervention is worth sending, and deliver through the Signal Arbiter.

```
Step 1: watch – subscribe to brain_signals via Supabase Realtime OR run
Step 2: evaluate – compute whether an intervention candidate is worth cr
Step 3: arbitrate – write candidate to proactive_signals queue (Signal A
```

Agents do not store state themselves. All state lives in the brain. This means any agent can be replaced or upgraded without losing the student's history.

3.3 Phase 1 — Mock Brain (Build Now)

During Phase 1, agents use a local mock context object instead of calling the NeuroAGI brain API. This allows all agents to be built and tested immediately without waiting for the brain layer.

```
{
  "student_id": "mock_001",
  "learning_style": "visual",
  "knowledge_gaps": ["integration by parts", "organic mechanisms"],
  "stress_level": 0.6,
  "upcoming_deadlines": [
    { "course": "CHEM 201", "assignment": "Lab Report 3", "due": "2026-06-01" },
  ],
  "preferred_language": "en",
  "session_count": 14
}
```

3.4 Phase 2 — Live Brain (After Ryan's API is ready)

Replace the mock context with:

```
const context = await brain.read(student_id)
// ... agent logic ...
await brain.write(student_id, signal)
```

No other change to any agent is required.

3.5 Proactivity Infrastructure

This section defines the infrastructure that allows FschooAI to act on behalf of the student without waiting for them to open the app. It is the backbone of all background and proactive agents.

3.5.1 Trigger / Event Runtime

Two mechanisms fire background agents. Both must be implemented.

Event-driven (real-time): Supabase Realtime listens for `INSERT` events on the `brain_signals` table via `pg_notify`. When a new signal is written (e.g., Canvas Agent writes a `stress_signal` after detecting 3 deadlines in 48 hours), the event runtime fires the relevant background agents immediately. A grade posted at 3pm cannot wait until the 2am Reflection run — it must trigger the Intervention Agent within minutes.

Scheduler (cron): Time-based triggers for agents that need to run on a fixed schedule regardless of events. Examples: Reflection Agent at 2am nightly, Canvas sync every 6 hours, spaced-repetition reminders at the student's preferred study time.

```
// Event-driven trigger (Supabase Realtime)
supabase
  .channel('brain_signals')
  .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'brain_signals' }, (payload) => triggerRuntime.dispatch(payload))
  .subscribe()

// Cron trigger (example)
cron.schedule('0 2 * * *', () => reflectionAgent.runForAllStudents())
```

3.5.2 Signal Arbiter

The Signal Arbiter is the most important missing piece in a naive proactive system. Without it, multiple background agents fire simultaneously and the student receives a flood of notifications that they immediately disable.

How it works:

1. Every background agent that wants to reach the student writes a **candidate signal** to the `proactive_signals` queue — it does NOT send a notification directly.
2. The Arbiter is **triggered by writes to** `proactive_signals`, not by a polling cron. When a new candidate is written for a student, a 2–3 minute debounce window opens. Any additional

candidates for the same student written within that window are batched together. After the window closes, the Arbiter runs once for that student's batch.

3. For each student batch, it reads all pending candidates, then:

- **Deduplicates** — removes redundant candidates (e.g., two agents both flagging the same deadline)
- **Ranks** — scores each candidate by `urgency × value` . Urgency = time sensitivity. Value = estimated benefit to the student.
- **Enforces rate limits** — maximum 3 proactive messages per student per day. Maximum 1 per hour.
- **Enforces quiet hours** — no delivery between 11pm and 8am unless the student has overridden this.
- **Selects** — approves the top-ranked candidate(s) and writes them to `notification_queue` .

4. Rejected candidates are discarded or deferred to the next cycle.

5. A **low-frequency safety sweep cron** (once per hour) scans for candidates whose `expires_at` has passed without being processed — this handles edge cases where the event trigger was missed. It does NOT process all students on every run.

The Arbiter is the confidence gate. Nothing reaches the student without passing through it.

Implementation note: The debounce is implemented as a short-lived lock per `student_id` in Redis or Supabase. On the first write for a student, set a lock with a 2-minute TTL and schedule the Arbiter run. Subsequent writes within the TTL extend the window by 1 minute (max 3 minutes total). This prevents the all-students sweep that a polling cron would require at scale.

3.5.3 Delivery Layer

Approved notifications in `notification_queue` are delivered through one of four channels based on urgency and student preference:

Channel	When to use	Service
In-app banner	Student is active in the app	Supabase Realtime push to frontend
Push notification	Student has app installed, not currently active	Firebase Cloud Messaging (FCM)
SMS	High-urgency, student not reachable by push	Twilio SMS API
Email	Low-urgency summaries, weekly reports	Resend or SendGrid

Delivery tracking: Every notification records `delivered_at`, `opened_at`, and `action_taken` (did the student act on it?). This data feeds the effectiveness feedback loop.

Quiet hours: Configurable per student. Default: no delivery 11pm–8am. SMS is never sent during quiet hours regardless of urgency.

3.5.4 Effectiveness Feedback Loop

The hard-coded thresholds in the Intervention Agent (stress > 0.8, 3+ sessions, etc.) are starting values only. They must be tuned per student over time.

Mechanism:

- Every `intervention_accepted` signal (student engaged with the notification) is a positive label.
- Every `intervention_delivered` with no action within 2 hours is a negative label.
- After 20 labelled examples per student, the system adjusts that student's thresholds: if they consistently ignore stress-level interventions but respond to deadline interventions, the stress threshold is raised and the deadline threshold is lowered.
- Per-channel tuning: if a student never opens push notifications but always responds to SMS, the delivery layer learns to prefer SMS for that student.

3.5.5 Cold-Start Mode

On Day 1, the brain is empty. Behavioural triggers (stress level, confusion patterns, session history) have no data to fire on. The system must not be silent on Day 1.

Degraded mode (Day 1 through Day 7):

- **Deadline-based proactivity is available immediately** — Canvas data is synced at signup. The Intervention Agent can fire deadline reminders from the first hour.
- **Behavioural proactivity is gated** — stress level, confusion detection, and pattern-based triggers are disabled until a baseline exists (minimum: 5 sessions + 7 days of data).
- **Learning style proactivity is gated** — adaptive explanation format is set to a neutral default until the learning style assessment is complete.

The UI must not show empty states as errors. During cold-start, show: “I’m learning how you work. The more you use FschooAI, the more personalised I become.”

3.6 Graph-Dependent Behaviours — Gated on Live NeuroAGI Brain

Critical distinction: The `StudentBrain` schema in §3.1 describes a typed knowledge graph with `knowledge_nodes`, prerequisite edges, and confidence scores. The Phase 1 mock brain (§3.3) and the `brain_context` Supabase table are flat JSON — a `knowledge_gaps` array and a `stress_level` float. These are not the same thing.

The following behaviours **require the live NeuroAGI graph brain** and are **not buildable in Phase 1** with the flat mock:

Behaviour	Why it needs the graph	Phase 1 fallback
Prerequisite checking (“do they understand the product rule before integration by parts?”)	Requires typed prerequisite edges between knowledge nodes	Tutor Agent checks for prerequisite by asking the student directly: “Before I explain this, do you know X?”
“Prerequisite mastered > 0.85” positive trigger (Intervention Agent)	Requires per-node confidence scores from the graph	Disable this trigger in Phase 1. Remove from the Intervention Agent’s trigger table until the live brain is available.
Knowledge node decay (concepts not revisited in 30+ days lose confidence)	Requires per-node confidence scores and last-reviewed timestamps	Reflection Agent skips decay in Phase 1. Flat mock does not track per-concept recency.
Cross-course knowledge graph visualisation (Max tier)	Requires the full graph structure	Defer to Phase 2. Show a placeholder in the Max tier dashboard.

Schedule warning: §8 lists “Replace mock brain with live NeuroAGI API” as a Week 6 task. This is aspirational, not a firm dependency. The NeuroAGI brain is an active research stack — the graph layer, multi-hop traversal, and confidence scoring are not guaranteed to be production-ready on a fixed date. All agents must be designed to function with the flat mock indefinitely. The Phase 2 brain swap is a progressive enhancement, not a hard launch blocker.

For Ryan: The flat mock is the contract. Any behaviour that requires more than `learning_style`, `knowledge_gaps[]`, `stress_level`, `upcoming_deadlines[]`, `preferred_language`, and `session_count` must be explicitly flagged as graph-dependent and gated. Do not design agent logic that silently breaks when the graph is unavailable.

4. Agent Specifications

Agent 1 — Reggie (Orchestrator)

Owner: Vincent

Environment: All — Reggie is the entry point for every student interaction

Priority: P0 — must be built first, all other agents depend on it

What it does: Reggie receives the student’s message, reads the brain context, decides which specialist agent to call, passes the message and context to that agent, and returns the response to the student. Reggie is the router and the face of FschooAI.

Input: Student message (text, voice, image)

Output: Response from the appropriate specialist agent

Routing logic:

Student message type	Routes to
“Explain this concept” / “I don’t understand X”	Tutor Agent
“What do I have due?” / “Show my schedule”	Canvas Agent + Planner Agent
“Help me study for my exam”	Exam Mode Agent
“Make a study plan for this week”	Planner Agent
“Summarise this lecture”	Lecture Agent
“Find me resources on X”	Library Agent
“I’m stressed / overwhelmed”	Intervention Agent
“Translate this”	Audio Agent
Ambiguous	Reggie asks one clarifying question, then routes

Brain signals written after each session:

- `session_type` : what kind of help was requested
- `session_duration` : how long the session lasted
- `topics_discussed` : list of subjects covered

Agent 2 — Canvas Agent

Owner: Aryan

Environment: Canvas LMS integration

Priority: P0 — foundational data source for all other agents

What it does: Connects to the student’s Canvas account via OAuth. Syncs all courses, assignments, due dates, grades, and syllabus documents. Stores this data in FschooAI’s Supabase database. Alerts the student about upcoming deadlines and missing work.

Canvas data pulled:

- All enrolled courses (current semester)
- All assignments per course (title, due date, points, submission status)

- All grades (current score, letter grade, course average)
- Syllabus documents (PDF/HTML)
- Announcement feed
- Professor contact information

Sync schedule: Every 6 hours automatically. On-demand when student opens FschooAI.

Alerts generated:

- Assignment due in less than 24 hours — push notification
- Assignment due in less than 72 hours — in-app banner
- Grade posted — in-app notification
- Missing assignment — weekly summary

Brain signals written:

- `course_list` : current courses
- `upcoming_deadlines` : next 14 days of assignments
- `grade_summary` : current standing per course
- `stress_signal` : if 3+ assignments due within 48 hours → `stress_level` += 0.2

Syllabus lookahead (required): After syncing the syllabus, the Canvas Agent must extract the weekly topic schedule from the syllabus document and store it as structured data in `course_content`. This enables the Planner Agent and Tutor Agent to know what topics are coming up in the next 1–2 weeks — not just what is due, but what will be taught. The Intervention Agent can then fire a proactive trigger: “You have stereochemistry coming up in CHEM 201 next week and your brain shows a gap there. Want to review it now?” Without syllabus lookahead, proactive preparation is impossible.

Bring-your-own past material: Students who transfer from another institution or who have past course materials (notes, old exams, textbooks) can upload them to FschooAI. The Canvas Agent is not responsible for this — it is a separate upload flow in the UI. Uploaded materials are stored in `course_content` with `source = 'student_upload'` and are available to the Tutor Agent and Lesson Generator as grounding context. This is a Phase 1 feature — do not defer it to Phase 2.

Note for Aryan: Canvas uses OAuth 2.0. The student authorises FschooAI to read their Canvas data. FschooAI never stores the Canvas password. The Canvas API token is stored encrypted in Supabase.

Agent 3 — Tutor Agent

Owner: Tencent engineer

Environment: Chat / Study Session

Priority: P0 — core product, primary reason students pay

What it does: This is the heart of FschooAI. The Tutor Agent answers academic questions, explains concepts, identifies what the student does not understand, and adapts its explanation style to the student's learning profile. It does not just answer — it teaches.

Key behaviours:

Adaptive explanation: The Tutor Agent reads the student's learning style from the brain and adjusts its response format accordingly.

Learning style	Response format
Visual	Diagrams described in text, tables, step-by-step with visual structure
Auditory	Conversational, narrative, “imagine you are explaining to a friend”
Reading	Dense text, definitions, citations, structured paragraphs
Kinesthetic	Examples first, then theory, “try this yourself” prompts

Knowledge gap detection: When the student asks a question, the Tutor Agent identifies whether the question reveals a deeper gap. If the student asks “why does integration by parts work?”, the agent checks whether they understand the product rule first. If not, it teaches the prerequisite before answering the original question.

Socratic mode: For questions that are clearly graded homework or exam questions, the Tutor Agent does not give the answer directly. It asks guiding questions that lead the student to the answer themselves. **This is NOT configurable for graded assignments** — the student cannot turn off Socratic mode for questions that are identifiable as graded work (i.e., the question matches an open assignment in Canvas with a future due date). Socratic mode can be turned off only for practice problems, past assignments, and conceptual questions that are not tied to a graded deliverable. This is an academic integrity guardrail, not a preference setting.

Simplification levels: Content must be adjustable to the student's level. International students and students who are not native English speakers need simpler language. The agent detects this from the student's communication style and adjusts automatically.

Input: Student question (text or image of problem)

Output: Explanation, worked example, follow-up question, or resource recommendation

Brain signals written:

- `topic_studied` : subject and concept
- `confusion_detected` : boolean — did the student express confusion?
- `gap_identified` : specific knowledge gap found
- `session_quality` : inferred from follow-up questions and engagement
- `explanation_format_used` : which format was used, for future optimisation

Agent 4 — Planner Agent

Owner: Bytedance engineer

Environment: Study Planner / Dashboard

Priority: P1

What it does: Builds personalised weekly study schedules based on the student's deadlines, available time, subject difficulty, and current stress level. The plan is not generic — it is specific to this student's brain context.

Planning inputs (all from brain context):

- Upcoming deadlines (from Canvas Agent)
- Current stress level
- Knowledge gaps (which subjects need more time)
- Historical study patterns (when does this student actually study?)
- Student's stated available hours

Plan output format:

```
Monday June 23:
  9:00-10:30  CHEM 201 — Stereochemistry review (gap identified)
  14:00-15:00 MATH 204 — Practice integration by parts (gap identified)
  19:00-20:00 Essay outline for ENGL 301 (due June 26)

Tuesday June 24:
  ...
```

Adaptive replanning: If the student misses a study block, the Planner Agent detects this (no activity during scheduled time) and automatically reschedules the missed work into the next available slot. It does not nag — it silently adjusts.

Brain signals written:

- `study_plan_created` : date and week covered
 - `plan_adherence` : percentage of planned blocks completed
 - `reschedule_count` : how many times the plan was adjusted
-

Agent 5 — Lecture Agent

Owner: Aryan

Environment: Chrome Extension / In-class

Priority: P1

What it does: Captures lecture audio through the Chrome extension, transcribes it in real time, generates a structured summary, identifies key concepts, and creates flashcards and quiz questions from the lecture content. The student gets a complete lecture package within minutes of the lecture ending.

Lecture package output:

- Full transcript (searchable)
- Summary (3–5 key points)
- Concept list (terms defined)
- Flashcard set (auto-generated)
- 5 quiz questions (multiple choice + short answer)
- Connections to existing brain knowledge (“This concept relates to what you studied last week in CHEM 201”)

Note for Aryan: The Chrome extension captures audio from the student’s microphone (in-class) or from a browser tab (online lecture). Audio is processed via Whisper. The transcript and summary are stored in Supabase and linked to the student’s brain.

Brain signals written:

- `lecture_attended` : course, date, duration
 - `concepts_introduced` : list of new concepts from this lecture
 - `lecture_summary_generated` : boolean
-

Agent 6 — Library Agent

Owner: Bytedance engineer

Environment: Resource Library

Priority: P2

What it does: When the student needs resources — textbook chapters, academic papers, YouTube explanations, practice problems — the Library Agent finds and curates them. It does not return a generic Google search. It returns resources matched to the student's learning style, current knowledge level, and specific gap.

Search behaviour:

- Reads the student's learning style from brain context
- Reads the specific knowledge gap being addressed
- Returns 3–5 resources ranked by relevance and quality
- Includes a one-sentence explanation of why each resource was chosen

Resource types:

- Khan Academy videos (visual learners)
- Academic papers (reading learners)
- Practice problem sets (kinesthetic learners)
- Podcast episodes / audio explanations (auditory learners)
- YouTube explanations (all types)

Brain signals written:

- `resource_recommended` : resource type and topic
 - `resource_clicked` : whether the student opened the resource
-

Agent 6b — Lesson Generator (Brain-Grounded Video)

Owner: Aryan (pipeline) + Tencent engineer (script generation)

Environment: Max tier only — gated at \$20/month

Priority: P2 (Phase 1 Max tier launch feature)

What it does: Generates a short personalised video lesson (2–4 minutes) for a specific concept the student is struggling with. This is not a generic explainer video. The script is built entirely from the student's brain context — their specific knowledge gap, their lecture transcript for the relevant course, their syllabus, and the professor's terminology. "A video built from your CHEM

201 lecture and the exact step you got wrong twice” is the product. A generic explainer is a commodity.

Brain-grounded script generation (required, not optional): The script generator must use all of the following as input context:

- The student’s specific knowledge gap (from `brain_context.knowledge_gaps`)
- The lecture transcript for the relevant course (from Lecture Agent output)
- The course syllabus (from Canvas Agent)
- The professor’s terminology and examples (extracted from lecture transcript)
- The student’s learning style (from `brain_context.learning_style`)
- The specific question or problem the student got wrong (from `session_history`)

A script generated without this context is not acceptable. If the lecture transcript is not available, the Lesson Generator must prompt the student to record a lecture first before generating the video.

Pipeline (async):

```
Step 1: Script generation – GPT-4o/Claude Sonnet, brain-grounded (30–60 seconds)
Step 2: Animation/visual generation – Manim or similar (60–120 seconds)
Step 3: TTS voiceover – ElevenLabs (10–20 seconds)
Step 4: Render and stitch (30–60 seconds)
Step 5: Deliver via notification_queue: "Your CHEM 201 stereochemistry video is ready"
```

Async delivery (required): Video generation takes 2–5 minutes end-to-end. This breaks the 3-second NFR in §9. Video generation is explicitly exempt from the 3-second NFR. The student triggers generation and receives a notification when the video is ready. The UI shows a progress indicator. Latency budget: < 5 minutes from trigger to delivery notification.

Segment-level regeneration: The pipeline must be chunked so a single segment can be re-run without rebuilding the whole video. The decision for Phase 1 is: **pre-generated branching** (cheaper, faster to build). The script is divided into 3–5 segments. Each segment is rendered independently. When the student requests a change (“explain this part differently”), only the relevant segment is re-generated and re-rendered. True segment regeneration (re-running the full pipeline for one segment) is the Phase 2 upgrade.

Cohort amortisation (cost control): When the Cohort Agent detects that 10+ students in the same course section are confused about the same concept, it triggers a shared video generation job. The core asset (script + animation + voiceover) is generated once. The

personalisation layer (intro referencing the student's specific mistake, outro referencing their next deadline) is generated per-student. This reduces the per-video cost from ~1.00 to 0.15 for cohort-triggered videos. The Lesson Generator must support a `cohort_mode: true` flag that separates the core asset generation from the personalisation layer.

Brain signals written:

- `video_generated` : concept, course, duration
 - `video_watched` : boolean, percentage watched
 - `video_segment_regenerated` : which segment was re-run
-

Agent 7 — Exam Mode Agent

Owner: Vincent

Environment: Exam Preparation

Priority: P1

What it does: When the student has an exam coming up, Exam Mode Agent creates a personalised exam preparation plan. It identifies the highest-priority topics based on the student's knowledge gaps, generates practice questions, simulates exam conditions, and tracks progress toward exam readiness.

Exam prep flow:

1. Student says "I have an exam in CHEM 201 in 3 days"
2. Agent reads brain context: knowledge gaps, past quiz performance, time available
3. Agent generates a 3-day prep plan prioritising the student's weakest areas
4. Agent generates 20 practice questions (mix of difficulty levels)
5. Student completes practice questions — agent evaluates answers
6. Agent updates the plan based on performance: if student struggles with topic X, add more practice on X
7. Day before exam: final review summary — "You are strong on A and B. Focus your last hour on C."

Brain signals written:

- `exam_prep_started` : course and exam date
 - `practice_questions_completed` : count and score
 - `exam_readiness_score` : 0–100 estimated readiness
-

- `weak_areas_at_exam_time` : final gap list before exam

Agent 8 — Intervention Agent

Owner: Vincent

Environment: Background monitor

Priority: P1

Agent pattern: Pattern B (Watch/Arbitrate/Deliver). This agent does not respond to user input — it watches `brain_signals` and writes candidates to the Signal Arbiter.

What it does: Runs silently in the background. Monitors the student's stress signals, deadline proximity, and engagement patterns. When it detects a student who is overwhelmed, falling behind, or disengaged, it intervenes proactively. It also fires on positive opportunity triggers — not just problems.

Negative intervention triggers (problems):

Signal	Threshold	Intervention
Stress level	> 0.8	"You have 3 assignments due in 48 hours. Want me to build a plan?"
No study activity	3+ days before deadline	"Your essay is due in 3 days and I haven't seen you work on it. Want to start now?"
Repeated confusion on same topic	3+ sessions	"You've asked about integration by parts 3 times. Let me try a different explanation."
Grade drop	> 10% below course average	"Your CHEM 201 grade dropped this week. Want to review what was covered?"
Late night pattern	Study sessions after 1am for 3+ nights	"You've been studying late. A 20-minute review now is more effective than 2 hours at midnight."

Positive opportunity triggers (advancement):

Signal	Condition	Intervention
Free study block + quiz tomorrow	Calendar gap detected + assignment due < 24h	"You have 90 free minutes at 3pm and a quiz tomorrow. Want to do a quick review now?"
Prerequisite mastered	Brain confidence score on prerequisite > 0.85	"You've got the product rule down. Ready to tackle integration by parts?"
Spaced-repetition due	Concept last reviewed > 7 days ago + exam within 14 days	"It's been 8 days since you reviewed stereochemistry. A 10-minute refresher now will stick better than cramming."
Streak opportunity	Student studied 4 days in a row	"4-day streak. One more session today and you'll have your best week this semester."

Stress level cap and escalation path (required): The Intervention Agent must not escalate indefinitely. If a student's stress level exceeds 0.9 for 3+ consecutive days and the student has not engaged with any intervention, the agent must:

1. Stop sending stress-related notifications (the student is clearly not responding — more notifications make it worse)
2. Show a single in-app message on next open: "It looks like you're going through a tough week. FschooAI is here when you're ready. Here are some campus mental health resources."
3. Write `stress_escalated: true` to the brain and suppress all stress-triggered notifications for 48 hours
4. After 48 hours, reset to normal monitoring

The agent must never imply a clinical diagnosis, never use the word "anxiety" or "depression", and never suggest the student is failing. Language must be supportive and non-judgmental. The campus mental health resource link is configurable per institution.

Important: All triggers — positive and negative — write to the `proactive_signals` queue. The Signal Arbiter (§3.5.2) decides what actually reaches the student. This agent does not send notifications directly.

Brain signals written:

- `intervention_triggered` : type and trigger (positive or negative)
- `intervention_accepted` : boolean — did the student engage with the intervention?

Agent 9 — Audio Agent

Owner: Aryan

Environment: Audio / Multilingual

Priority: P2

What it does: Handles all audio-related tasks. Transcribes lecture recordings, translates content into the student's preferred language, converts text explanations to audio for auditory learners, and processes voice input from the student.

Capabilities:

- Speech-to-text (Whisper) — transcribe any audio file or live recording
- Text-to-speech — read explanations aloud
- Translation — translate lecture content, study materials, or AI responses into the student's language
- Language detection — automatically detects the student's preferred language from their messages

Supported languages (Phase 1): English, Mandarin Chinese, Hindi, French, Spanish, Arabic

Brain signals written:

- `preferred_language` : detected from student's messages
- `audio_sessions` : count of audio-mode interactions

Agent 10 — Office Hours Agent

Owner: Tencent engineer

Environment: Office Hours Preparation

Priority: P2

Academic integrity guardrail — writing feedback only: When a student asks for help with a written assignment (essay, report, lab write-up), this agent and the Tutor Agent must provide **feedback and suggestions only — not rewritten text**. Specifically:

- The agent may identify structural weaknesses: "Your thesis statement is unclear — it does not state a position."
- The agent may suggest what to improve: "Your second paragraph lacks a topic sentence."
- The agent may NOT rewrite sentences, paragraphs, or sections for the student.
- The agent may NOT generate a full outline and then write content for each section.

- The agent may generate a **blank structural scaffold** (section headings only, no content) to help the student organise their own writing.

This guardrail applies to all written assignments tied to an open Canvas assignment with a future due date. For past assignments or practice writing, the restriction is lifted.

What it does: Helps the student prepare for and follow up on professor office hours. Before office hours, it generates a list of specific questions based on the student's knowledge gaps. After office hours, it helps the student record what was discussed and updates the brain with new knowledge.

Pre-office hours flow:

1. Student says "I have office hours with Professor Chen in 30 minutes"
2. Agent reads brain context: current gaps in the relevant course
3. Agent generates 3–5 specific, well-formed questions to ask the professor
4. Agent briefs the student: "Here is what you should ask and why"

Post-office hours flow:

1. Student says "Office hours just ended"
2. Agent asks: "What did you learn? What was clarified?"
3. Student describes what happened
4. Agent updates the brain: gaps closed, new concepts added

Brain signals written:

- `office_hours_attended` : course and professor
- `questions_prepared` : list of questions generated
- `gaps_closed` : knowledge gaps resolved in the session

Agent 11 — Calendar Agent

Owner: Bytedance engineer

Environment: Calendar Integration

Priority: P2

What it does: Integrates with Google Calendar and Apple Calendar. Reads the student's existing schedule (classes, work, commitments) and uses this to make the Planner Agent's study plans realistic. It also writes study blocks back to the student's calendar.

Calendar reads:

- Class schedule (recurring events)
- Work shifts
- Social commitments
- Existing study blocks

Calendar writes:

- Study blocks generated by Planner Agent
- Exam dates (from Canvas)
- Assignment deadlines (from Canvas)

Brain signals written:

- `available_hours_per_day` : calculated from calendar gaps
 - `calendar_connected` : boolean
-

Agent 12 — Reflection Agent

Owner: Ryan (NeuroAGI)

Environment: Background — runs nightly

Priority: P1

What it does: This agent runs every night at 2am for each student. It reviews the day's interactions, synthesises patterns, updates the brain graph, decays stale knowledge nodes, and prepares the brain for the next day. This is the agent that makes the brain smarter over time.

Nightly tasks:

- Review all signals written by other agents during the day
- Update knowledge gap confidence scores (did the student show improvement?)
- Decay old knowledge nodes (concepts not revisited in 30+ days lose confidence)
- Identify new patterns (e.g., “student consistently struggles on Mondays after weekend”)
- Update stress level based on weekly trajectory
- Generate a “brain health summary” for the student (optional, shown in dashboard)

Note: This agent lives inside NeuroAGI, not FschooIAI. Ryan owns it. FschooIAI agents do not call it directly — it runs automatically on the NeuroAGI brain layer.

Agent 13 — Terminal Agent

Owner: Vincent

Environment: Developer / Power User Mode

Priority: P3

What it does: For advanced students who want to query their own brain directly, run custom workflows, or inspect their data. Think of it as a command-line interface to the student's brain.

Example commands:

```
> show my knowledge gaps in CHEM 201
> what topics have I studied this week?
> export my study history as CSV
> set my learning style to visual
> show my stress level trend this month
```

Brain signals written:

- `terminal_commands_used` : list of commands executed

Agent 14 — Cohort / Collective Intelligence Agent

Owner: Ryan (NeuroAGI) + Vincent (FschoolAI integration)

Environment: Background — runs on event trigger (new confusion signals) and nightly

Priority: P2 — requires canonical entity layer and k-anonymity minimum before activation

Agent pattern: Pattern B (Watch/Arbitrate/Deliver). This agent is a **producer into the Signal Arbiter** — its outputs fan out to each cohort member's Intervention Agent and Planner Agent. It does not communicate with students directly.

What it does: Aggregates anonymised, de-identified learning signals across students in the same course section. Identifies concept gaps that are widespread in the cohort. Surfaces targeted review recommendations to each individual student based on what their cohort is struggling with collectively.

The core insight: If 15 students in one section hit confusion on the same concept this week, that is a *leading* signal available immediately — grades are lagging, sparse, and privacy-sensitive. Confusion clustering is the right signal to build on first.

What it does NOT do:

- It does not aggregate grades or grade distributions (privacy-hot, consent-gated, Phase 3 only)
- It does not make claims about professor performance — never “your prof’s test was unfair”
- It does not show individual student data to other students — ever
- It does not operate on cohorts smaller than 10 students (k-anonymity minimum)

Canonical entity layer (required prerequisite): Today the `courses` table is keyed by `student_id`, so the same Canvas course is N separate rows with no way to aggregate across them. Before this agent can function, a canonical entity layer must be built:

```
-- Canonical course – shared across all students in the same Canvas instance
canonical_courses (
  id                uuid PRIMARY KEY,
  institution_id    text,                -- e.g., "carleton.ca"
  canvas_course_id  text,                -- Canvas's own course ID (shared across students)
  course_name       text,
  semester          text,
  UNIQUE(institution_id, canvas_course_id)
)

-- Canonical assignment – shared across all students in the same course
canonical_assignments (
  id                uuid PRIMARY KEY,
  canonical_course_id  uuid REFERENCES canonical_courses(id),
  canvas_assignment_id text,
  title             text,
  due_date           timestamp,
  UNIQUE(canonical_course_id, canvas_assignment_id)
)
```

Cohorts are grouped by `(institution_id, canvas_course_id)`. These IDs are shared across students in the same Canvas instance.

Concept taxonomy (required prerequisite — harder than the canonical course layer): The confusion clustering algorithm joins on `concept_tag`. If tags are free-text strings written by the Tutor Agent at inference time, the same concept will appear as “stereochem”, “stereochemistry”, “chirality”, “R/S configuration”, and “chiral centres” — none of which will reach the k=10 threshold individually.

Two approaches are acceptable. Choose one before building the Cohort Agent:

Option A — Canonical concept ontology per subject: A curated taxonomy of concept tags per subject domain (e.g., Organic Chemistry: `[stereochemistry, reaction_mechanisms, functional_groups, ...]`). The Tutor Agent maps its free-text gap identification to the nearest canonical tag at write time. Requires upfront curation per subject. Recommended for Phase 1 subjects (STEM, economics).

Option B — Embedding-based tag normalisation: The Tutor Agent writes a free-text concept description. At aggregation time, the Cohort Agent embeds all concept tags and clusters them by cosine similarity (threshold: 0.85). Concepts within the same cluster are merged into a representative tag. No upfront curation, but requires an embedding model call per aggregation run.

Both options must be decided and built before the Cohort Agent is activated. Add the chosen approach to the build order in §8 under Week 7+.

Confusion clustering algorithm:

1. Every time the Tutor Agent writes a `confusion_detected` signal, it includes `canonical_course_id` and `concept_tag`.
2. The Cohort Agent aggregates these signals per `(canonical_course_id, concept_tag)` over a rolling 7-day window.
3. If 10+ students in the same cohort show confusion on the same concept within 7 days, a cohort insight is generated.
4. The insight is written to the `proactive_signals` queue for each cohort member: “15 students in your CHEM 201 section are struggling with stereochemistry this week. Here is a targeted 10-minute review.”

Privacy architecture (non-negotiable):

- All cohort aggregation runs against a **de-identified aggregation store** — a separate table with RLS policies that prevent any per-student data from being exposed.
- **k-anonymity minimum: 10 students.** If a cohort has fewer than 10 students, no insight is computed or shown. If a concept has fewer than 10 confused students, no insight is generated.
- **Per-student consent flag:** Students must opt in to cohort intelligence. Default is opt-out. The consent flag is stored in `students.cohort_consent` `boolean DEFAULT false`.
- **Legal review required:** This feature requires reconciling §9 (data belongs to student, never aggregated without consent) and §11 (social features deferred). Legal review for PIPEDA and FERPA compliance is required before this agent goes live. Do not ship without legal sign-off.

Framing rule: Insights are always framed as concept-gap recommendations for the individual student, never as commentary on the professor or course quality.

Correct framing	Prohibited framing
“Many students in your section are finding stereochemistry difficult. Here’s a targeted review.”	“Your professor didn’t explain this well.”
“This concept is commonly misunderstood in CHEM 201. Let me break it down differently.”	“Your class average on this topic is low.”

Brain signals written (to de-identified aggregation store only):

- `cohort_insight_generated` : concept tag, cohort size, confusion count
- `cohort_insight_delivered` : how many students received the insight

5. User Flows

5.1 Onboarding Flow

The onboarding is the “identity card” session — a one-on-one setup with FschooIAI that builds the student’s initial brain profile.

Step 1 — Account creation Student signs up with email or Google. A blank brain object is created with their `student_id`.

Step 2 — Canvas connection Student connects their Canvas account via OAuth. Canvas Agent syncs all courses, assignments, and grades. The brain is populated with course context and upcoming deadlines.

Step 3 — Learning style assessment Reggie asks 5 quick questions to determine the student’s learning style. These are conversational, not a formal test. Example: “When you are trying to understand something new, do you prefer to see a diagram, hear an explanation, read about it, or try it yourself?”

Step 4 — First brain summary FschooIAI shows the student their initial brain: “Here is what I know about you so far. You are taking 4 courses. Your next deadline is in 2 days. I think you learn best visually. Is this right?”

Step 5 — First interaction Reggie asks: “What do you want to work on today?” The student is now in the product.

5.2 Daily Use Flow

Student opens FschooIAI

- Reggie shows a personalised daily brief:
"Good morning. You have 2 assignments due this week.
You have a study block for CHEM 201 at 2pm.
Your exam readiness for MATH 204 is 62% – want to practice?"

Student asks a question **or** picks a task

- Reggie routes **to** the appropriate agent
- Agent **reads** brain context
- Agent responds
- Agent writes signal **to** brain

End of session

- Reflection Agent (nightly) synthesises the **day**
- Brain **is** updated
- Tomorrow's **brief is prepared**

5.3 Exam Preparation Flow

Student: "I have a CHEM 201 exam on Friday"

- Reggie routes **to** Exam Mode Agent
- Exam Mode Agent reads brain: gaps **in** stereochemistry **and** reaction mechanisms
- Agent creates **3**-day prep plan
- Day **1**: stereochemistry practice (student's weakest area)
- Day **2**: reaction mechanisms + mixed practice
- Day **3**: full mock exam + review
- **Each** day: agent evaluates student's answers, updates readiness score
- Thursday night: "You are at 78% readiness. Focus on reaction mechanisms"

6. Data Model

6.1 Core Tables (Supabase)

students

```
id          uuid PRIMARY KEY
email       text UNIQUE NOT NULL
name        text
canvas_token text (encrypted)
created_at  timestamp
last_active timestamp
```

brain_context (Phase 1 mock — later replaced by NeuroAGI API)

```
id          uuid PRIMARY KEY
student_id  uuid REFERENCES students(id)
learning_style text
stress_level float
knowledge_gaps jsonb
preferences jsonb
updated_at  timestamp
```

courses

```
id          uuid PRIMARY KEY
student_id  uuid REFERENCES students(id)
canvas_course_id text
name        text
professor   text
semester    text
```

assignments

```
id                uuid PRIMARY KEY
course_id         uuid REFERENCES courses(id)
canvas_assignment_id text
title             text
due_date          timestamp
points_possible   float
points_earned      float
submitted         boolean
```

sessions

```
id                uuid PRIMARY KEY
student_id        uuid REFERENCES students(id)
agent_used        text
input_text        text
output_text       text
topics            text[]
duration_seconds   int
created_at        timestamp
```

brain_signals

```
id                uuid PRIMARY KEY
student_id        uuid REFERENCES students(id)
signal_type       text
signal_data       jsonb
created_at        timestamp
```

proactive_signals (candidate interventions awaiting Signal Arbiter decision)

```

id                uuid PRIMARY KEY
student_id        uuid REFERENCES students(id)
agent_source       text                -- which agent wrote this candidate
urgency_score      float                -- 0.0-1.0, time sensitivity
value_score        float                -- 0.0-1.0, estimated benefit
message_text       text                -- the message to deliver if approved
channel_hint       text                -- preferred channel (push, sms, email,
status            text DEFAULT 'pending' -- pending | approved | rejected
created_at         timestamp
expires_at         timestamp           -- candidate is discarded after this ti

```

notification_queue (approved interventions ready for delivery)

```

id                uuid PRIMARY KEY
student_id        uuid REFERENCES students(id)
proactive_signal_id uuid REFERENCES proactive_signals(id)
channel           text                -- actual delivery channel chosen b
message_text       text
scheduled_for      timestamp          -- when to deliver (respects quiet l
delivered_at       timestamp
opened_at         timestamp
action_taken       boolean            -- did the student act on it?
created_at        timestamp

```

canonical_courses (shared across students in the same Canvas instance)

```

id                uuid PRIMARY KEY
institution_id     text                -- e.g., "carleton.ca"
canvas_course_id  text                -- Canvas's own course ID
course_name       text
semester         text
UNIQUE(institution_id, canvas_course_id)

```

canonical_assignments (shared across students in the same course)

```

id                uuid PRIMARY KEY
canonical_course_id  uuid REFERENCES canonical_courses(id)
canvas_assignment_id text
title             text
due_date          timestamp
UNIQUE(canonical_course_id, canvas_assignment_id)

```

cohort_confusion_signals (de-identified aggregation store — separate RLS, no per-student data)

```

id                uuid PRIMARY KEY
canonical_course_id  uuid REFERENCES canonical_courses(id)
concept_tag       text
confusion_count    int           -- number of students confused
week_start        date
updated_at        timestamp

```

flashcard_reviews (SRS state — required for spaced-repetition trigger)

```

id                uuid PRIMARY KEY
student_id        uuid REFERENCES students(id)
flashcard_id      uuid           -- references the flashcard generator
concept_tag       text
ease_factor       float DEFAULT 2.5 -- FSRS ease factor
interval_days     int  DEFAULT 1   -- current review interval in days
repetitions       int  DEFAULT 0   -- number of times reviewed
next_review_at    timestamp       -- when this card is next due
last_reviewed_at  timestamp
rating           int             -- last review rating: 1 (again) 2
created_at        timestamp

```

SRS engine: The spaced-repetition scheduling uses the **FSRS algorithm** (Free Spaced Repetition Scheduler — open-source, more accurate than SM-2). It runs client-side in the browser/app. On each flashcard review, the student rates their recall (1–4). The FSRS algorithm updates `ease_factor`, `interval_days`, and `next_review_at`. The “spaced-repetition due” trigger in the Intervention Agent reads `next_review_at` to determine when to send a

reminder. **This table and scheduling engine must be built before the spaced-repetition trigger is activated.** Remove the trigger from the Intervention Agent’s positive trigger table until `flashcard_reviews` is populated with real data.

7. Technical Stack

Layer	Technology
Frontend	React 19, Tailwind CSS 4, shadcn/ui
Mobile	React Native (Phase 2)
Backend	Supabase (database + auth + storage)
Brain layer	NeuroAGI API (Phase 2) / local mock JSON (Phase 1)
LLM	GPT-4o via OpenAI API (Phase 1)
Speech-to-text	OpenAI Whisper
Canvas integration	Canvas LMS REST API + OAuth 2.0
Calendar integration	Google Calendar API + Apple CalDAV
Chrome extension	Manifest V3, React
Hosting	Manus (FschoolAI frontend)
Notification delivery	Firebase Cloud Messaging (push), Twilio (SMS), Resend (email)
SRS engine	FSRS algorithm (open-source, client-side)
Stripe	Payment processing (Phase 1 launch requirement)

7.1 Cost Envelope and Model Routing

This section is a launch blocker. With 14 agents, nightly Reflection, cohort aggregation, an arbiter, and video generation, the default “use GPT-4o for everything” approach is not economically viable. The following routing rules are required before launch.

Per-active-user cost target: $< 1.50/month$ for a Free user, $< 3.00/month$ for a Pro user, $< \$5.00/month$ for a Max user. Gross margin targets: Free (0% — acquisition cost), Pro (~75%),

Max (~70% including video).

Model routing rules:

Task	Model	Rationale
Intent classification (Reggie routing)	GPT-4o-mini or Claude Haiku	Sub-100ms, < \$0.001/call, runs on every message
Signal evaluation (Intervention Agent, Arbiter scoring)	GPT-4o-mini	Structured JSON output, no long context needed
Summarisation (Lecture Agent summary, Planner output)	GPT-4o-mini	High volume, quality threshold is moderate
Flashcard and quiz generation	GPT-4o-mini	Template-driven, low creativity requirement
Tutoring (Tutor Agent, Exam Mode)	GPT-4o or Claude Sonnet	Core product, quality is the moat — do not downgrade
Video script generation (Lesson Generator)	GPT-4o or Claude Sonnet	Max-tier feature, brain-grounded, quality matters
Reflection synthesis (Reflection Agent)	GPT-4o-mini	Runs nightly, structured signal processing
Cohort insight generation	GPT-4o-mini	Templated output, low creativity

Prompt caching: The brain context object (`learning_style` , `knowledge_gaps` , `stress_level` , `upcoming_deadlines`) is stable within a session. Cache it as a system prompt prefix using OpenAI's prompt caching feature. Estimated saving: 40–60% of token cost for multi-turn sessions.

Cost per active user per day (estimated at launch):

User type	Sessions/day	Est. LLM cost/day	Est. LLM cost/month
Free (casual, 20 msg cap)	1 session, ~15 messages	\$0.02	\$0.60
Pro (regular, unlimited)	2–3 sessions, ~40 messages	\$0.06	\$1.80
Max (power user, video)	3+ sessions + 1 video/week	\$0.12	\$3.60

These are estimates based on GPT-4o-mini for routing/evaluation and GPT-4o for tutoring at current API pricing. Validate against actual usage in the first 30 days and adjust routing rules accordingly.

Video cost note (Max tier): A single brain-grounded video (script + ElevenLabs TTS + animation render) costs approximately 0.80–1.20 *per video*. At 4 videos/month per Max user, this is 3.20–4.80/month in video costs alone. Cohort amortisation (see Agent 14) reduces this when the same core video is reused across cohort members with only the framing layer personalised.

8. Build Order and Ownership

The build order is designed so no engineer blocks another. Ryan’s brain mock is available from Day 1 so all agents can develop against it.

Week	What gets built	Owner
Week 1	Brain mock JSON + <code>brain.read()</code> / <code>brain.write()</code> stub	Ryan
Week 1	Canvas Agent — OAuth + sync	Aryan
Week 1	Reggie — basic routing	Vincent
Week 1	Supabase schema — all tables including <code>proactive_signals</code> , <code>notification_queue</code> , <code>canonical_courses</code>	Ryan
Week 2	Tutor Agent — core explanation logic	Tencent engineer
Week 2	Planner Agent — study schedule generation	Bytedance engineer
Week 2	Lecture Agent — transcription + summary	Aryan
Week 2	Trigger / Event Runtime — Supabase Realtime + cron scaffold	Ryan
Week 3	Exam Mode Agent	Vincent
Week 3	Intervention Agent (Pattern B, writes to Arbiter)	Vincent
Week 3	Library Agent	Bytedance engineer
Week 3	Signal Arbiter — dedup, rank, rate-limit, quiet hours	Ryan
Week 3	Delivery Layer — FCM push + Twilio SMS + email (Resend) + in-app	Aryan
Week 4	Audio Agent	Aryan

Week	What gets built	Owner
Week 4	Office Hours Agent	Tencent engineer
Week 4	Calendar Agent	Bytedance engineer
Week 4	Effectiveness feedback loop — per-student threshold tuning	Ryan
Week 5	Reflection Agent (NeuroAGI)	Ryan
Week 5	Terminal Agent	Vincent
Week 5	Cold-start mode — deadline-only proactivity until baseline exists	Vincent
Week 6	Replace mock brain with live NeuroAGI API	Ryan + all
Week 7+	Cohort Agent — requires canonical layer + 10+ students + legal sign-off	Ryan + Vincent
Week 7+	Canonical entity layer (<code>canonical_courses</code> , <code>canonical_assignments</code>)	Ryan

9. Non-Functional Requirements

Response time: Every agent must respond within 3 seconds for standard queries. Lecture transcription and exam prep plan generation may take up to 10 seconds with a loading indicator.

Privacy: Student data is never used to train models. Canvas tokens are stored encrypted. Brain data belongs to the student — they can export or delete it at any time.

Language: All agents must support English by default. Mandarin Chinese support is required for Phase 1 (Chinese student market). Additional languages via Audio Agent in Phase 2.

Offline mode: Core brain context is cached locally. Students can view their study plan and upcoming deadlines without internet. Active agent sessions require internet.

Accessibility: All UI must meet WCAG 2.1 AA. Voice input must be available for all agent interactions.

10. Success Metrics

Metric	Target (Month 3)
Daily active users	500+
Session length	> 8 minutes average
Canvas connections	> 70% of users connect Canvas
Agent interactions per session	> 2 agents used per session
Student-reported grade improvement	> 40% of users report improvement
Retention (Day 30)	> 45%
NPS	> 50
Free → Pro conversion rate	> 8% within 30 days of signup
Pro → Max upgrade rate	> 15% of Pro users within 60 days
LLM cost per active Pro user/month	< \$3.00
LLM cost per active Max user/month	< \$5.00
Proactive notification open rate	> 35%
Proactive notification disable rate	< 5% (measures Arbiter quality)
Video completion rate (Lesson Generator)	> 60% of generated videos watched to completion

11. Out of Scope (Phase 1)

The following are explicitly out of scope for Phase 1 and should not be built until Phase 2:

- NeuroAGI hardware integration (Neural Card)

- School/institution-facing dashboard
- Professor tools
- Native mobile app (iOS/Android)
- Social features (study groups, leaderboard)
- EducAI integration

Payment processing is IN scope for Phase 1. FschooIAI launches with a Free tier and two paid tiers:

Tier	Price	Key Phase 1 features
Free	\$0	20 messages/day, basic brain, Canvas sync
Pro	\$12/month	Unlimited chat, nightly reflection, proactive interventions, Lesson Generator (10/month), Exam Predictor
Max	\$20/month	Everything in Pro + unlimited Lesson Generator, Brain export, Brain API access, cross-course knowledge graph

Stripe integration is required before public launch. The Lesson Generator (video generation feature) is a **Max-tier feature** — it is gated behind \$20/month and must not be accessible to Free or Pro users. See §7.1 for cost envelope and model routing. See TOKEN_ECONOMY.md for full tier feature breakdown.

Video generation (Lesson Generator) is Phase 1 Max-tier only. It is not a generic explainer — see Agent 6 (Library Agent) and the Lesson Generator spec for the brain-grounded video pipeline.

This document is the source of truth for FschooIAI Phase 1 engineering. Any questions, contact Vincent Yang.